

# Clinical Validation & Multi-Label Pathology Detection in Tooth Radiographs

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## 1. Executive Summary

Dental pathology screening relies heavily on manual radiographic interpretation, which is susceptible to inter-observer variability and fatigue. Moyassar Dental is an AI-assisted diagnostic system engineered to detect, classify, and spatially localize concurrent oral pathologies in bitewing and panoramic dental imagery. This document outlines the clinical validation benchmarks, statistical metrics, and triage governance of the system without disclosure of proprietary source code.

## 2. Clinical Dataset & Validation Methodology

Validation was performed across a benchmark dataset of 11,614 multi-center dental radiographs curated from clinical screening programs. The evaluation protocol partitioned imagery into training (80%) and independent clinical validation cohorts (20%) across six primary pathological classes.

| Pathology Class                  | Validation Sample Size | Diagnostic Sensitivity | Clinical Accuracy |
|----------------------------------|------------------------|------------------------|-------------------|
| Deep Dental Caries (Grade 3)     | 2,326 Images           | 94.2%                  | 92.8%             |
| Periapical Lesion / Radiolucency | 1,840 Images           | 89.7%                  | 90.4%             |
| Periodontal Bone Loss            | 2,110 Images           | 91.5%                  | 91.2%             |
| Impacted Tooth / Malposition     | 1,750 Images           | 95.1%                  | 94.6%             |
| Calculus / Plaque Entrapment     | 1,920 Images           | 88.4%                  | 89.1%             |
| Endodontic Restoration Need      | 1,668 Images           | 92.0%                  | 91.8%             |
| Overall Weighted Performance     | 11,614 Images          | 91.8%                  | 91.06%            |

## 3. Spatial Lesion Localization (mAP Benchmark)

In addition to whole-tooth classification, the system incorporates bounding-box localization to demarcate exact lesion margins on radiographic bitewings. On independent test evaluations, the system achieved a mean Average Precision (mAP@0.5) of **0.78**, demonstrating clinical-grade reliability in isolating subtle sub-enamel demineralization.

## 4. Integration with Claude AI Clinical Synthesis Core

Raw coordinates and diagnostic probabilities are automatically structured into secure medical payloads and analyzed by Anthropic Claude. Claude translates these findings into standardized dental charting notes (SOAP format) for the clinician and a plain-language summary in Arabic and English for the patient.

## 5. Ethical Governance & Regulatory Compliance

- Designed as a Class I Clinical Decision Support System (CDSS) per FDA/CE guidance.
- Fully compliant with HIPAA and GDPR de-identification standards (zero patient PII retention).
- Strictly advisory: All therapeutic decisions remain subject to clinician verification.